AERO 151

Advanced Drone Regulations and Safety Management
Course structure 2 units, 1 lecture and 1 lab. Enrollment cap 25. Small lab.
Semester hours 18 lecture, 54 lab, 36 outside work, and 0 activity. 108 total student hours.
Minimum qualifications Aeronautics

Course description

This two-unit lecture and laboratory course develops students' ability to navigate FAA regulatory processes and manage safety in professional drone operations beyond introductory Part 107 preparation. Students examine waivers, airspace authorizations, coordination with aviation authorities, and the supporting documentation needed to explain how a proposed operation can be conducted safely. Topics include operations beyond visual line of sight, Part 108 rulemaking, and other emerging regulatory frameworks, with attention to distinguishing proposed changes from requirements currently in effect. Safety management instruction addresses organizational responsibilities, hazard identification, risk assessment, reporting, and evaluation of whether safety measures are working. Through case studies and applied laboratory scenarios, students connect regulatory requirements with practical operating procedures and defensible safety decisions.

Course rationale

Drone regulations continue to evolve, and some enterprise operations require FAA permissions or waivers beyond routine flight operations. Advanced pilots and drone program managers need the ability to evaluate these requirements and manage organizational safety. This course supports professionals responsible for complex operations, including utility work, through advanced regulatory practice and safety management aligned with FAA standards.

Connection to the College Mission

The course supports students' career goals by preparing them for greater responsibility in professional drone programs. It develops skills that can strengthen job performance, career advancement, and earning potential while supporting responsible operations.

Program fit

The course does not fit within an existing degree or certificate. It will support proposed certificates of specialization. Transfer or articulation is not proposed.

Related courses and consultation

AERO 100 introduces drone regulations and safety. AERO 151 extends that foundation to advanced permissions, waivers, and organizational safety management. Consultation with related disciplines has established support for the proposed scope.

Enrollment and department efficiency

A cap of 25 supports individual feedback on regulatory analysis and safety decisions. The smaller section size allows substantive case discussion, supervised laboratory exercises, and review of students' reasoning.

Resources and funding

AERO 161

Drone Applications in Geospatial Technology
Course structure 3 units, 2 lecture and 1 lab. Enrollment cap 25. Small lab.
Semester hours 36 lecture, 54 lab, 72 outside work, and 0 activity. 162 total student hours.
Minimum qualifications Aeronautics

Course description

This course develops the skills needed to collect and prepare reliable spatial data using drones for a broad range of professional applications. Students learn the complete orthomosaic production process, from selecting camera settings and designing flight missions to processing overlapping imagery into a georeferenced image of an area. Field exercises incorporate GNSS receivers, high-accuracy positioning, ground control, and independent checkpoints to establish and evaluate positional accuracy. Instruction also introduces other sensors and data products, allowing students to consider collection methods beyond conventional photography and photogrammetry. A culminating project integrates safe mission planning, field operations, processing, quality checks, and documentation to deliver usable datasets for GIS and other professional applications.

Course rationale

Employers need usable, reliable spatial data, and drone operators need more than flight skills to produce it. This course prepares students for mapping work in surveying, construction, environmental monitoring, and infrastructure management. It strengthens employment prospects and helps working professionals add drone data collection to their existing skills. The course will support proposed certificates of specialization.

Connection to the College Mission

The course supports student success and career goals through practical skills in producing reliable spatial data. These skills can improve employment opportunities, job performance, and access to more advanced technical work.

Program fit

The course does not fit within an existing degree or certificate. It will support proposed certificates of specialization. Transfer or articulation is not proposed.

Related courses and consultation

Related content appears in GEOL 230, GEOL 233, GEOL 234, and ENGR 201. GIS, engineering, and surveying faculty have been consulted and support the course as a distinct, complementary offering. Its emphasis is drone mission design, aircraft and sensor operation, and production of reliable aerial datasets.

Enrollment and department efficiency

A cap of 25 matches AERO 100 and supports supervised fieldwork and individual assessment. Small crews allow students to rotate through flight, sensor, and data processing roles while receiving feedback.

Resources and funding

AERO 162

Drone Applications in Precision Agriculture
Course structure 3 units, 2 lecture and 1 lab. Enrollment cap 25. Small lab.
Semester hours 36 lecture, 54 lab, 72 outside work, and 0 activity. 162 total student hours.
Minimum qualifications Aeronautics

Course description

This course introduces drone applications in precision agriculture, using information and technology to support crop management based on conditions at particular locations within a field. Students collect visible and multispectral imagery to document crop variability, support scouting, and identify areas that warrant closer field investigation. Thermal imaging exercises examine temperature patterns associated with crop water stress and irrigation performance, including the conditions that affect interpretation. Agricultural spraying instruction focuses on safely planning and executing drone missions that deliver chemicals to their intended treatment areas through appropriate flight paths, equipment settings, and coverage. The spraying component addresses drone operation and application placement rather than the pesticide applicator knowledge and licensing curriculum taught through specialized agricultural training.

Course rationale

Drones are an expanding tool in agriculture, creating opportunities for workers who can apply the technology effectively. This course helps students compete for agricultural jobs and helps working professionals add useful capabilities to their existing roles. Specialized drone skills can support advancement and earning potential in farm operations and agricultural services. The course will support proposed certificates of specialization.

Connection to the College Mission

The course supports students' employment and advancement goals through practical agricultural technology skills. It prepares students to improve their work, pursue new responsibilities, and contribute to the region's agricultural economy.

Program fit

The course does not fit within an existing degree or certificate. It will support proposed certificates of specialization. Transfer or articulation is not proposed.

Related courses and consultation

Related concepts appear in AGPS 230, AGPS 242, and GEOL 234. Consultation with related disciplines has established support for the course. Instruction focuses on drone sensing and application operations. Pesticide applicator knowledge and licensing instruction belong in agriculture and are outside this course's scope.

Enrollment and department efficiency

A cap of 25 matches AERO 100 and supports supervised agricultural flight exercises. Small crews provide opportunities to practice aircraft operation, sensor use, and application placement with individual feedback.

Resources and funding

AERO 163

Drone Applications in Public Safety
Course structure 3 units, 2 lecture and 1 lab. Enrollment cap 25. Small lab.
Semester hours 36 lecture, 54 lab, 72 outside work, and 0 activity. 162 total student hours.
Minimum qualifications Aeronautics

Course description

This course prepares students to support public safety missions with drones in search and rescue, fire response, law enforcement, and emergency management. Students develop skills in visible and thermal imaging to search for missing people, assess incident conditions, and document affected areas. Practical exercises include photogrammetric reconstruction of simulated collision and crime scenes, giving students experience capturing imagery and producing spatial records for investigation and documentation. Additional scenarios develop search planning, disaster assessment, and live aerial observation skills while emphasizing crew coordination, communication with incident command, and safe operation in changing environments. Students examine the legal and regulatory frameworks specific to public safety drone operations, including emergency airspace access, agency operating authority, privacy protections, and the handling of imagery and evidence.

Course rationale

Public safety agencies increasingly use drones as a tool for response, assessment, and documentation. This course helps existing public safety professionals develop additional operational capabilities and helps students pursuing those careers build relevant technical skills. It supports improved job performance and opportunities for specialized assignments and advancement. The course will support proposed certificates of specialization and complement preparation for the underlying public safety roles.

Connection to the College Mission

The course advances students' career goals by developing technical skills relevant to public safety employment and professional growth. Students learn to improve operational work while serving community needs and maintaining public trust.

Program fit

The course does not fit within an existing degree or certificate. It will support proposed certificates of specialization. Transfer or articulation is not proposed.

Related courses and consultation

Related content appears in CJ 210 and CJ 213, with additional connections to GIS. Consultation with related disciplines has established support for the course. Its distinct focus is drone operation, aerial data collection, and crew coordination in public safety scenarios.

Enrollment and department efficiency

A cap of 25 matches AERO 100 and supports supervised scenario exercises. Small crews allow individual assessment of flight, imaging, communication, and coordination skills while students rotate through operational roles.

Resources and funding

AERO 147 Proposed

Independent Studies in Drone Technology
Course structure 0.5 to 2 lab units. Enrollment cap 25.
Semester hours 0 lecture, 27 to 108 lab, 0 outside work, and 0 activity. 27 to 108 total student hours.
Minimum qualifications Aeronautics

Course description

This variable-unit course provides prepared students an opportunity for in-depth study or an applied project in drone technology under the direction of an instructor. Students build on completed preparatory coursework to investigate a topic beyond the scope of regularly scheduled classes. Projects may involve drone operations, sensors, data collection and analysis, equipment evaluation, or development of operational methods. The student and instructor establish project objectives, required work, progress meetings, and evaluation methods through an independent study agreement. Students document their work and present a final product, report, or demonstration appropriate to the project and units earned.

Course rationale

Prepared students need opportunities to pursue substantial projects that extend beyond scheduled coursework. Independent study allows them to develop specialized skills, solve practical problems, and produce work that demonstrates their capabilities to employers. A flexible project format supports individual career interests and professional growth without replacing existing courses.

Connection to the College Mission

The course supports student success through individual learning goals and applied work connected to employment or advancement. Students develop initiative, technical judgment, and evidence of their skills for future professional opportunities.

Program fit

The course does not fit within an existing degree or certificate. Transfer or articulation is not proposed.

Related courses and consultation

Projects build on AERO coursework and may connect with related technical disciplines. The course extends learning through individual projects and does not replace existing offerings. Project scope will be coordinated with relevant faculty. Consultation on the broader drone curriculum has established support from related disciplines.

Enrollment and department efficiency

A proposed cap of 25 retains the program’s enrollment limit. Individual project agreements define the scope of work and instructor consultation needed for each student. Enrollment and project assignments must remain compatible with effective supervision.

Resources and funding